

Linearization of an Affine Switched System

Consider the affine switched system

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, \quad x(t) \in \mathbb{R}^n. \quad (1)$$

The switching signal is piecewise constant. Its mode order is fixed, while its switching times may vary:

$$\sigma(t) = q_i = \bar{\sigma}(\bar{t}_{i-1}), \quad t_{i-1} \leq t < t_i, \quad i = 1, \dots, N. \quad (2)$$

Introduce the augmented state and matrix

$$\mathcal{X}(t) = \begin{bmatrix} x(t) \\ 1 \end{bmatrix}, \quad \tilde{A}_q = \begin{bmatrix} A_q & b_q \\ 0 & 0 \end{bmatrix}. \quad (1)$$

Then the affine system becomes the homogeneous system

$$\boxed{\dot{\mathcal{X}}(t) = \tilde{A}_{\sigma(t)}\mathcal{X}(t)}. \quad (2)$$

For each interval, define

$$\Delta t_i = t_i - t_{i-1}, \quad \Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1}, \quad \Delta t_i = \Delta \bar{t}_i + \delta t_i, \quad F_i = \tilde{A}_{q_i}. \quad (3)$$

The state transition on interval i is

$$\mathcal{X}(t_i) = e^{F_i \Delta t_i} \mathcal{X}(t_{i-1}). \quad (4)$$

Therefore,

$$\mathcal{X}(t_N) = e^{F_N \Delta t_N} e^{F_{N-1} \Delta t_{N-1}} \dots e^{F_1 \Delta t_1} \mathcal{X}(t_0). \quad (5)$$

The first interval is the rightmost factor.

Define the nominal transition matrix for interval i by

$$\phi_i = e^{F_i \Delta \bar{t}_i}. \quad (6)$$

Let

$$e(t) = \mathcal{X}(t) - \bar{\mathcal{X}}(t), \quad \mathcal{X}(t_0) = \bar{\mathcal{X}}(t_0) + e(t_0). \quad (7)$$

Using (5), the exact terminal error is

$$\begin{aligned} e(t_N) &= e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots \\ &\quad e^{F_1(\Delta \bar{t}_1 + \delta t_1)} (\bar{\mathcal{X}}(t_0) + e(t_0)) - \phi_N \phi_{N-1} \dots \phi_1 \bar{\mathcal{X}}(t_0). \end{aligned} \quad (8)$$

For a constant matrix F_i , expand $e^{F_i s}$ around $s = \Delta \bar{t}_i$:

$$\begin{aligned} \frac{d}{ds} e^{F_i s} &= F_i e^{F_i s}, \\ e^{F_i(\Delta \bar{t}_i + \delta t_i)} &= e^{F_i \Delta \bar{t}_i} + F_i e^{F_i \Delta \bar{t}_i} \delta t_i + \mathcal{O}(\delta t_i^2) \\ &= \phi_i + F_i \phi_i \delta t_i + \mathcal{O}(\delta t_i^2) \\ &\doteq \phi_i + F_i \phi_i \delta t_i. \end{aligned} \quad (9)$$

In the following, $\doteq$ denotes equality up to first order in the duration perturbations: terms containing two or more factors δt_i are discarded. Substituting the first-order approximation into every transition gives

$$\begin{aligned} &e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots e^{F_1(\Delta \bar{t}_1 + \delta t_1)} \\ &\doteq (\phi_N + F_N \phi_N \delta t_N)(\phi_{N-1} + F_{N-1} \phi_{N-1} \delta t_{N-1}) \dots (\phi_1 + F_1 \phi_1 \delta t_1) \\ &\doteq \phi_N \phi_{N-1} \dots \phi_1 \\ &\quad + F_N \phi_N \phi_{N-1} \dots \phi_1 \delta t_N \\ &\quad + \phi_N F_{N-1} \phi_{N-1} \dots \phi_1 \delta t_{N-1} \\ &\quad + \dots \\ &\quad + \phi_N \phi_{N-1} \dots \phi_2 F_1 \phi_1 \delta t_1. \end{aligned} \quad (10)$$

Substituting (10) into (8), cancelling the nominal term, and discarding $\delta t_i e(t_0)$ yields

$$\begin{aligned}
e(t_N) &\doteq \phi_N \phi_{N-1} \cdots \phi_1 e(t_0) \\
&\quad + \phi_N \phi_{N-1} \cdots \phi_2 F_1 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_1 \\
&\quad + \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \bar{\mathcal{X}}(t_0) \delta t_2 \\
&\quad + \cdots \\
&\quad + F_N \phi_N \phi_{N-1} \cdots \phi_1 \bar{\mathcal{X}}(t_0) \delta t_N.
\end{aligned} \tag{11}$$

Collect the duration perturbations and their sensitivity columns:

$$\begin{aligned}
\delta \mathbf{t} &= \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_N \end{bmatrix}, \\
\Gamma &= \begin{bmatrix} \phi_N \phi_{N-1} \cdots \phi_2 F_1 \phi_1 \bar{\mathcal{X}}(t_0), \\ \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \bar{\mathcal{X}}(t_0), \dots, \\ F_N \phi_N \phi_{N-1} \cdots \phi_1 \bar{\mathcal{X}}(t_0) \end{bmatrix}.
\end{aligned} \tag{12}$$

For the final compact form, define the complete nominal transition

$$\Phi = \phi_N \phi_{N-1} \cdots \phi_1. \tag{13}$$

Equation (11) is thus

$$\boxed{e(t_N) \doteq \Phi e(t_0) + \Gamma \delta \mathbf{t}}. \tag{14}$$